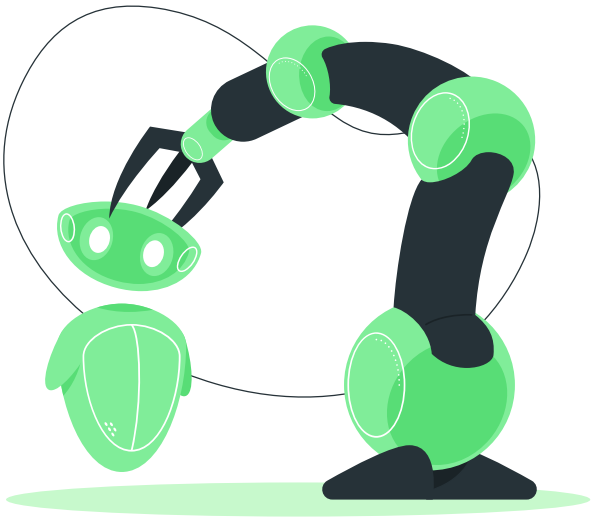
Robotics
CS 3015

Umarah Qaseem



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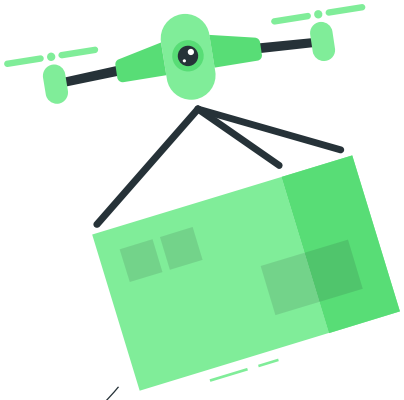
- Activity Quintic Polynomial
- Microcontroller vs Microprocessor
- ESP32,
- ESP8266 Introduction
- Specifications
- IDE Installation
- Setup and Configurations
- Hello World



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What did we learn?

- Measuring motion
- Inertial Navigation System
- Inertial Measurement Unit
- Inertial Sensors
- Accelerometer
- Magnetometers
- Gyroscope
- Gimbal Mechanisms
- Gimbal Lock
- Nasa's Apollo IMU
- Activity Quintic Polynomial



Revision

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Activity

-



Robotics – Umarah Qaseem

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Applying Boundary Conditions

$$s(t) = At^5 + Bt^4 + Ct^3 + Dt^2 + Et + F$$
$$\dot{s}(t) = 5At^4 + 4Bt^3 + 3Ct^2 + 2Dt + E$$
$$\ddot{s}(t) = 20At^3 + 12Bt^2 + 6Ct + 2D$$

time	s	$\dot{s}$	$\ddot{s}$
$t = 0$	s_0	$\dot{s}_0$	$\ddot{s}_0$
$t = T$	s_T	$\dot{s}_T$	$\ddot{s}_T$

when $t = 0$

$$s(0) = F$$
$$\dot{s}(0) = E$$
$$\ddot{s}(0) = 2D$$

when $t = T$

$$s(T) = AT^5 + BT^4 + CT^3 + DT^2 + ET + F$$
$$\dot{s}(T) = 5AT^4 + 4BT^3 + 3CT^2 + 2DT + E$$
$$\ddot{s}(T) = 20AT^3 + 12BT^2 + 6CT + 2D$$

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Solve for Coefficients

- Writing in matrix form

$$\begin{pmatrix} s_0 \\ s_T \\ \dot{s}_0 \\ \dot{s}_T \\ \ddot{s}_0 \\ \ddot{s}_T \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ T^5 & T^4 & T^3 & T^2 & T & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 5T^4 & 4T^3 & 3T^2 & 2T & 1 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 \\ 20T^3 & 12T^2 & 6T & 2 & 0 & 0 \end{pmatrix} \begin{pmatrix} A \\ B \\ C \\ D \\ E \\ F \end{pmatrix}$$

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Solve for Coefficients

- Invert the matrix to get coefficients

$$\begin{pmatrix} A \\ B \\ C \\ D \\ E \\ F \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ T^5 & T^4 & T^3 & T^2 & T & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 5T^4 & 4T^3 & 3T^2 & 2T & 1 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 \\ 20T^3 & 12T^2 & 6T & 2 & 0 & 0 \end{pmatrix}^{-1} \begin{pmatrix} s_0 \\ s_T \\ \dot{s}_0 \\ \dot{s}_T \\ \ddot{s}_0 \\ \ddot{s}_T \end{pmatrix}$$

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In terms of theta

$$\theta(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5$$

$$\begin{bmatrix} \theta_0 \\ \omega_0 \\ \alpha_0 \\ \theta_f \\ \omega_f \\ \alpha_f \end{bmatrix} = \begin{bmatrix} 1 & t_0 & t_0^2 & t_0^3 & t_0^4 & t_0^5 \\ 0 & 1 & 2t_0 & 3t_0^2 & 4t_0^3 & 5t_0^4 \\ 0 & 0 & 2 & 6t_0 & 12t_0^2 & 20t_0^3 \\ 1 & t_f & t_f^2 & t_f^3 & t_f^4 & t_f^5 \\ 0 & 1 & 2t_f & 3t_f^2 & 4t_f^3 & 5t_f^4 \\ 0 & 0 & 2 & 6t_f & 12t_f^2 & 20t_f^3 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \end{bmatrix}$$

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Example: Quintic Polynomial Trajectory

- Given: A robot joint moves from pi/3 to 7pi/4 rad over 4s, starting from rest and then stopping.
- Find: An appropriate quintic polynomial trajectory.

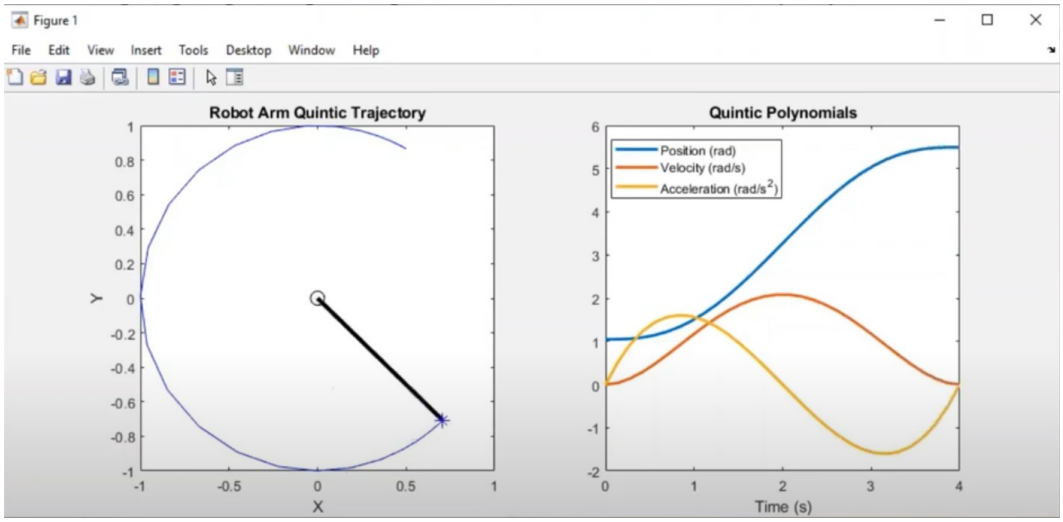
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Solution Steps – Summary

- Find q(0), v(0), a(0). Then q(T), v(T), a(T).
- Then write quintic polynomial equation of degree 5.
- Then take its single and double derivative.
- Now plug in the values.
- Convert this to matrix form of 6x6.
- Or remove 3 variables by plugging in their values and make it a 3x3 matrix, which easier to solve.
- Find C (matrix of coefficients). Then plug values of coeffecients back in q(t), v(t), a(t). You have got trajectories for all these parameters.

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Solution



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Easy Solution

- Put all the values and make it a 3x3 matrix!
- Hint: 3 unknowns with three equations. You can solve it.

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Microcontroller and Microprocessor



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Difference

- Microprocessor only consists of Central Processing Unit, whereas Microcontroller has memory, a CPU and I/O all integrated into one chip.
- A microprocessor uses external bus to interface to ROM, RAM, and other peripherals.

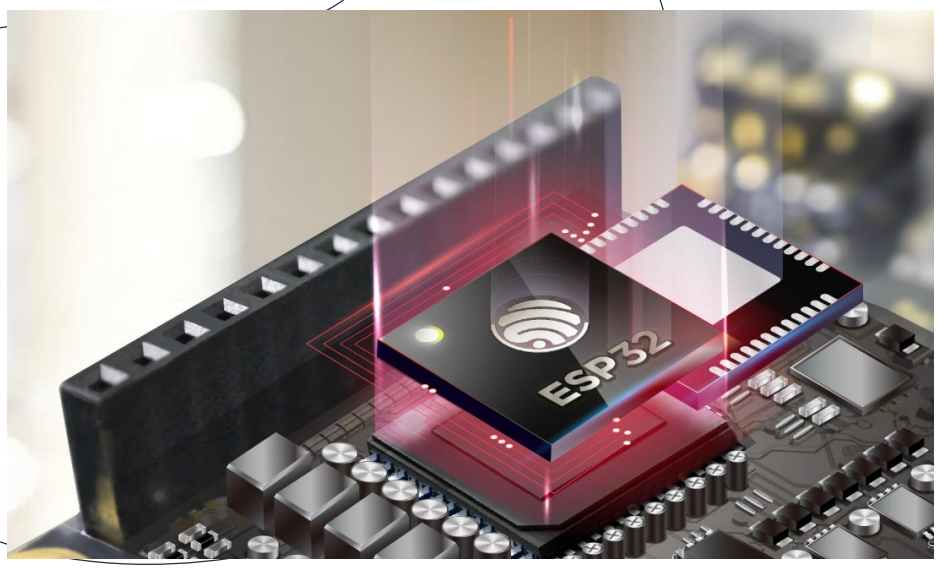
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Features

Features	Microcontrollers	Microprocessors
Integrations	Microcontrollers integrate CPU, memory and I/O ports onto a single chip.	Microprocessors require external components such as memory and additional peripherals.
Applications	Microcontrollers are better suited for specific, low-power or ultra-low power applications within embedded systems, such as home appliances like washing machines or IoT devices.	Microprocessors are more suitable for general-purpose and high-performance applications requiring more processing power, such as personal computers or the types of servers used in data centers.
Performance	Microcontrollers are optimized for efficiency and real-time processing and operate at lower clock speeds up to 200 MHz.	Microprocessors are built for more demanding and complicated computations and can operate at clock speeds above 1 GHz.
Cost	Microcontroller hardware is generally inexpensive and because they do not require specialized programming knowledge, they add very little to overall project budgets.	Microcontrollers are better suited for specific, low-power or ultra-low-power applications within embedded systems, such as home appliances like washing machines or IoT devices.

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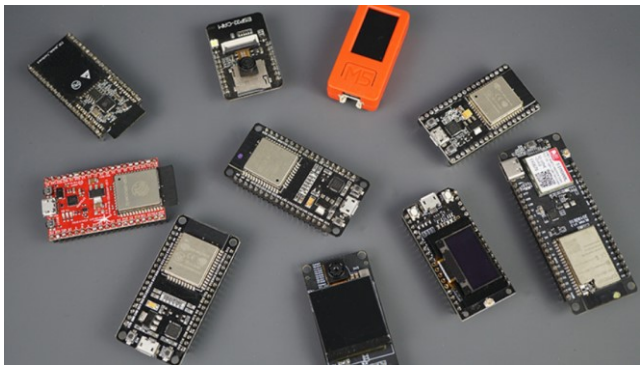
ESP32



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ESP32

- A feature-rich MCU with integrated Wi-Fi and Bluetooth connectivity for a wide-range of applications

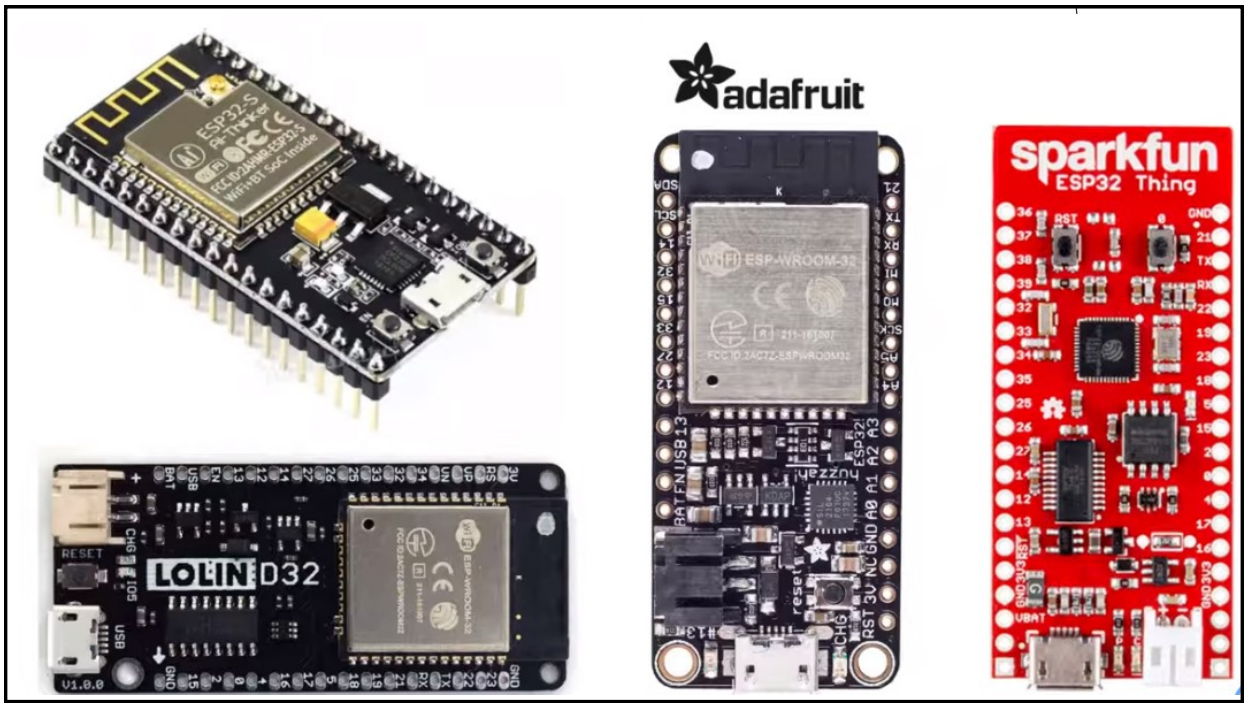


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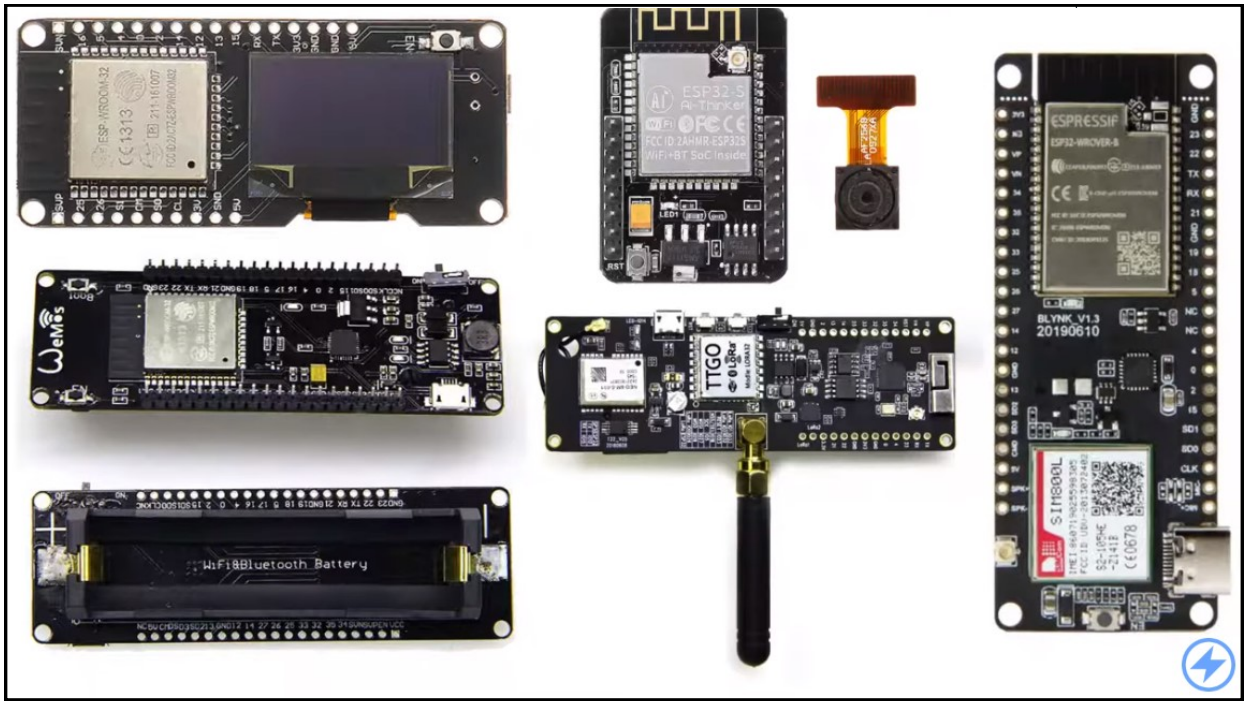
Specifications

- Integrated Crystal- 40 MHz
- Module Interfaces- UART, SPI, I2C, PWM, ADC, DAC, GPIO, pulse counter, capacitive touch sensor
- Integrated SPI flash- 4 MB
- ROM- 448 KB (for booting and core functions)
- SRAM- 520 KB
- Integrated Connectivity Protocols- WiFi, Bluetooth, BLE (BLE Support)
- On-chip sensor- Hall sensor
- Operating temperature range- -40 - 85 degrees Celsius
- Operating Voltage- 3.3V
- Operating Current- 80 mA (average)

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ESP32-S
Ai-Thinker
adafruit
sparkfun
ESP32 Thing
LOLIN D32
ESP-WROOM-32

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ESP32-S
Ai-Thinker
WeMos
TTGO
ESPRESSIF
BLYNK_V1.3
20190610
WiFi&Bluetooth Battery

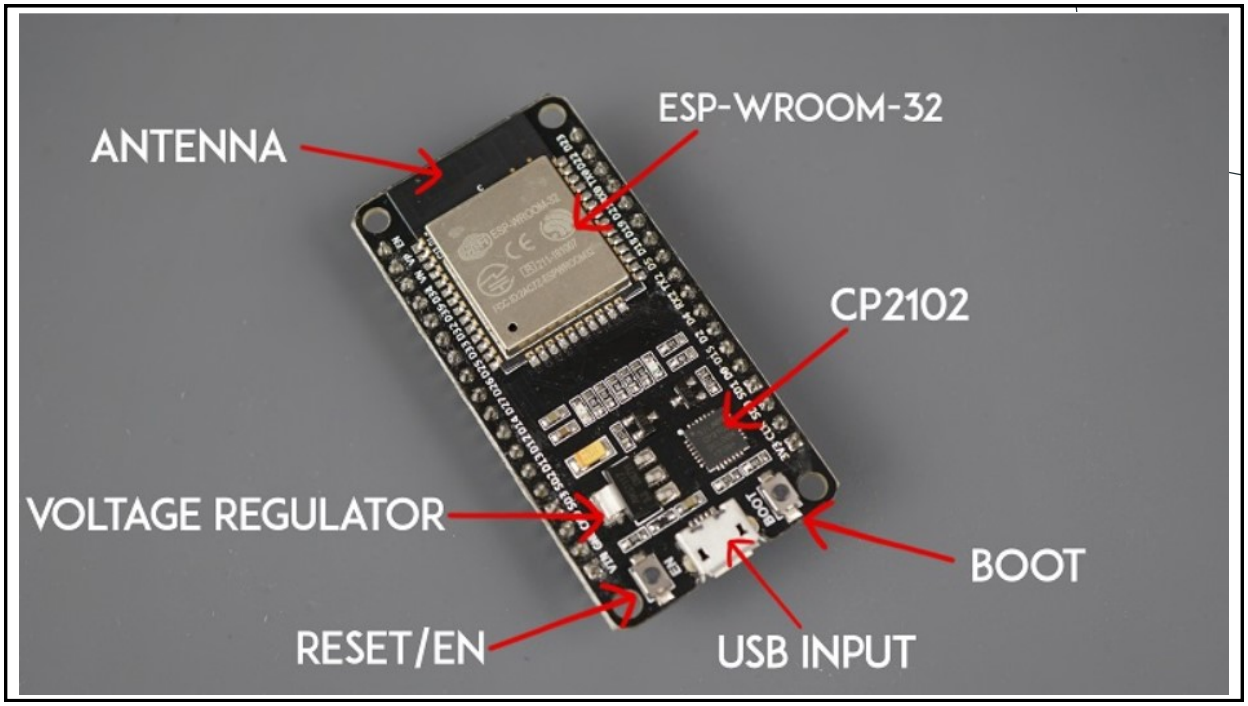
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Applications

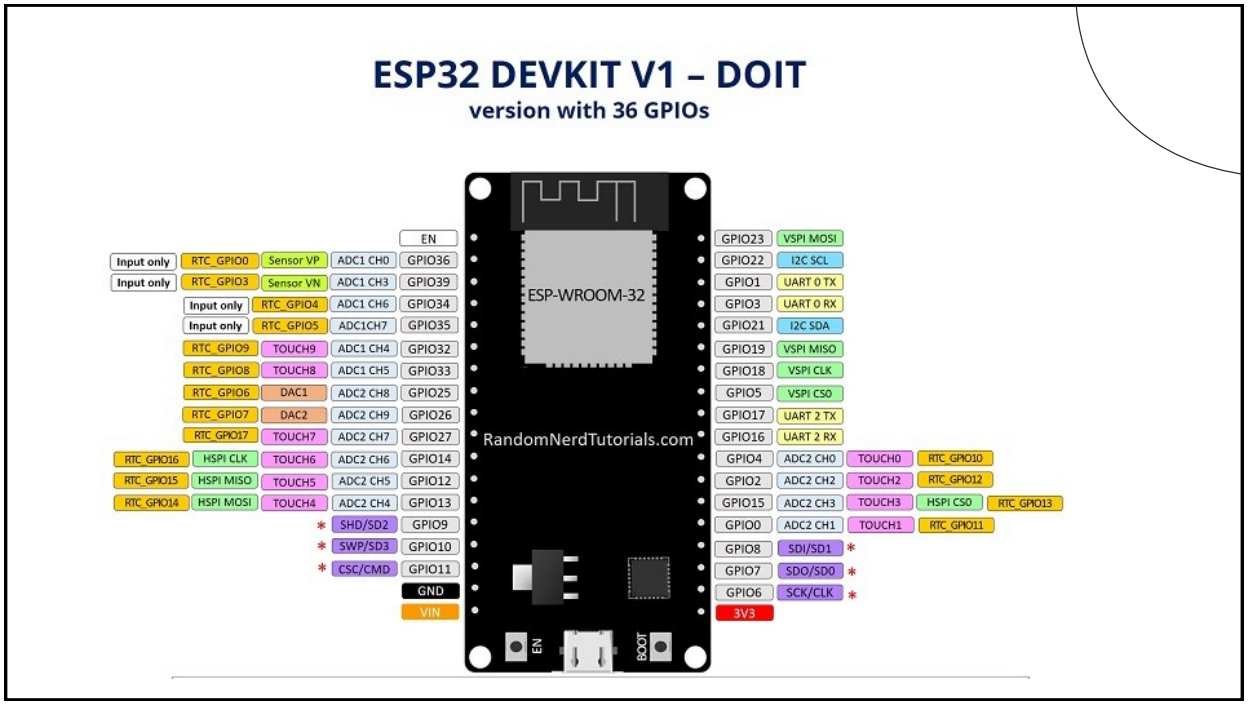
- Came around in 2016 as a successor to ESP8266.
- IoT devices
- Home automation
- Wireless communication
- Robotics
- Wearable devices
- Industrial automation
- Environmental monitoring
- Education and prototyping
- Security Systems
- Healthcare

Advantages

- Low-cost
- Low-power
- Wi-Fi capabilities
- Bluetooth
- Dual-core
- Rich peripheral input/output interface
- Compatible with the Arduino “programming language”
- Compatible with MicroPython


ANTENNA
ESP-WROOM-32
CP2102
VOLTAGE REGULATOR
BOOT
RESET/EN
USB INPUT

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ESP32 DEVKIT V1 – DOIT
version with 36 GPIOs
ESP-WROOM-32
RandomNerdTutorials.com
EN
Input only RTC_GPIO0 Sensor VP ADC1 CH0 GPIO36
Input only RTC_GPIO3 Sensor VN ADC1 CH3 GPIO39
Input only RTC_GPIO4 ADC1 CH6 GPIO34
Input only RTC_GPIO5 ADC1CH7 GPIO35
RTC_GPIO9 TOUCH9 ADC1 CH4 GPIO32
RTC_GPIO8 TOUCH8 ADC1 CH5 GPIO33
RTC_GPIO6 DAC1 ADC2 CH8 GPIO25
RTC_GPIO7 DAC2 ADC2 CH9 GPIO26
RTC_GPIO17 TOUCH7 ADC2 CH7 GPIO27
RTC_GPIO16 HSPI CLK TOUCH6 ADC2 CH6 GPIO14
RTC_GPIO15 HSPI MISO TOUCH5 ADC2 CH5 GPIO12
RTC_GPIO14 HSPI MOSI TOUCH4 ADC2 CH4 GPIO13
* SHD/SD2 GPIO9
* SWP/SD3 GPIO10
* CSC/CMD GPIO11
GND
VIN
GPIO23 VSPI MOSI
GPIO22 I2C SCL
GPIO1 UART 0 TX
GPIO3 UART 0 RX
GPIO21 I2C SDA
GPIO19 VSPI MISO
GPIO18 VSPI CLK
GPIO5 VSPI CS0
GPIO17 UART 2 TX
GPIO16 UART 2 RX
GPIO4 ADC2 CH0 TOUCH0 RTC_GPIO10
GPIO2 ADC2 CH2 TOUCH2 RTC_GPIO12
GPIO15 ADC2 CH3 TOUCH3 HSPI CS0 RTC_GPIO13
GPIO0 ADC2 CH1 TOUCH1 RTC_GPIO11
GPIO8 SDI/SD1 *
GPIO7 SDO/SD0 *
GPIO6 SCK/CLK *
3V3
EN
BOOT

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ESP32 based Game console

- Announced in 2019
- Released in 2022
- The SDK is compatible with both C and Lua programming languages.
- Uses ESP32 as its chip.



Industrial Example

- Why not you?
- You can make something like this too,
- or even better!



Programming languages, frameworks, platforms, and environments used for ESP32 programming:

- Arduino IDE with the ESP32 Arduino Core
- Espressif IoT Development Framework (IDF) – official Espressif development framework for ESP32
- MicroPython A lean implementation of Python 3 for microcontrollers
- Espressif Mesh Development Framework
- Espruino – JavaScript SDK and firmware closely emulating Node.js
- Lua RTOS for ESP32
- Moddable SDK — includes JavaScript language and library support for the ESP32
- Mongoose OS – an operating system for connected products on microcontrollers; programmable with JavaScript or C. A recommended platform by Espressif Systems,[39] AWS IoT,[40] and Google Cloud IoT.[41]
- mruby for the ESP32
- .NET nanoFramework - Coding in .NET C#, deploy and debug from Visual Studio[42]
- NodeMCU – Lua-based firmware
- PlatformIO Ecosystem and IDE
- Pymakr IDE – IDE designed for use with Pycom devices; handles firmware upgrades and includes MicroPython REPL console
- Simba Embedded Programming Platform
- Whitecat Ecosystem Blockly Based Web IDE
- Zerynth – Python for IoT and microcontrollers, including the ESP32

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More details

- For more details, official website:
- https://www.espressif.com/en/products/socs/esp32

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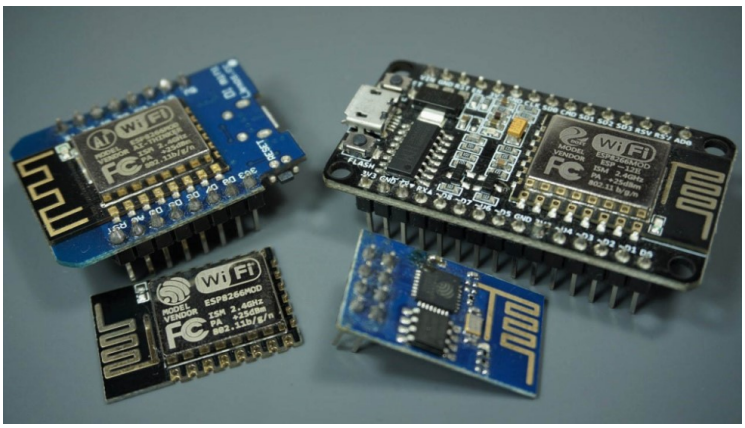
ESP8266

Why will we use this?



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Many versions



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Variants

- We will use ESP8266 NODEMCU V1.0 ESP-12E MODULE
- It has multiple variants

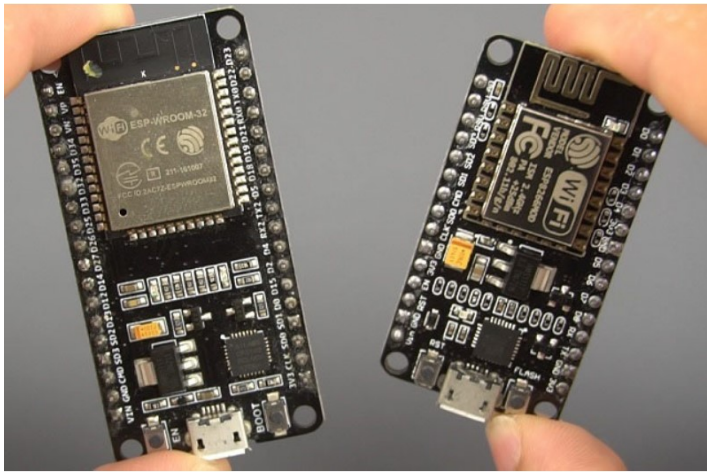
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Advantages

- Low-cost
- Low-power
- Wi-Fi capabilities
- Many peripheral input/output interface
- Compatible with the Arduino “programming language”
- Compatible with MicroPython

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Difference



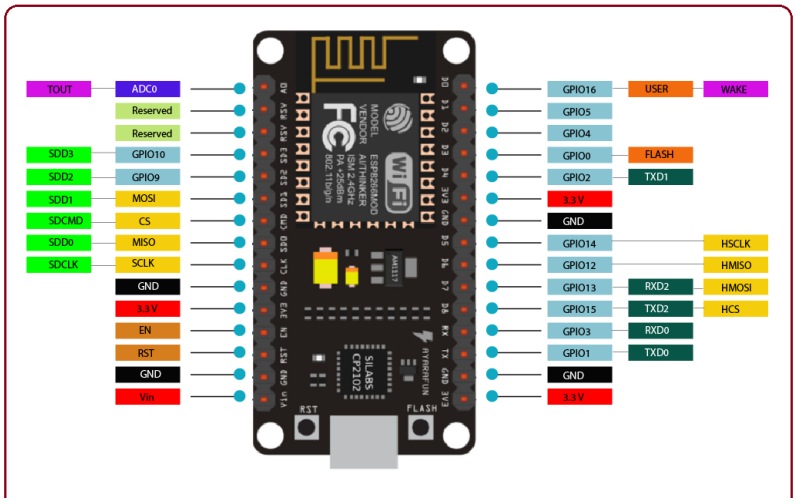
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Specifications

- Microcontroller: ESP-8266 32-bit
- Clock Speed: 80 MHz
- USB Converter: CP2102
- USB Connector: Micro USB
- Operating Voltage: 3.3V
- Flash Memory: 4 MB
- Digital I/O: 11
- Analog Inputs: 1
- Communications: Serial, SPI. I2C and 1-Wire via software libraries
- WiFi: Built-in 802.11 b/g/n

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Datasheet



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ESP32 vs ESP8266

	ESP8266	ESP32
MCU	Xtensa Single-core 32-bit L106	Xtensa Dual-Core 32-bit LX6 with 600 DMIPS
802.11 b/g/n Wi-Fi	HT20	HT40
Bluetooth	X	Bluetooth 4.2 and BLE
Typical Frequency	80 MHz	160 MHz
SRAM	X	✓
Flash	X	✓
GPIO	17	34
Hardware /Software PWM	None / 8 channels	None / 16 channels
SPI/I2C/I2S/UART	2/1/2/2	4/2/2/2
ADC	10-bit	12-bit
CAN	X	✓
Ethernet MAC Interface	X	✓
Touch Sensor	X	✓
Temperature Sensor	X	✓(old versions)
Hall effect sensor	X	✓
Price	$ (3$ – $6)	$$ ($6 – $12)

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IDE Installation



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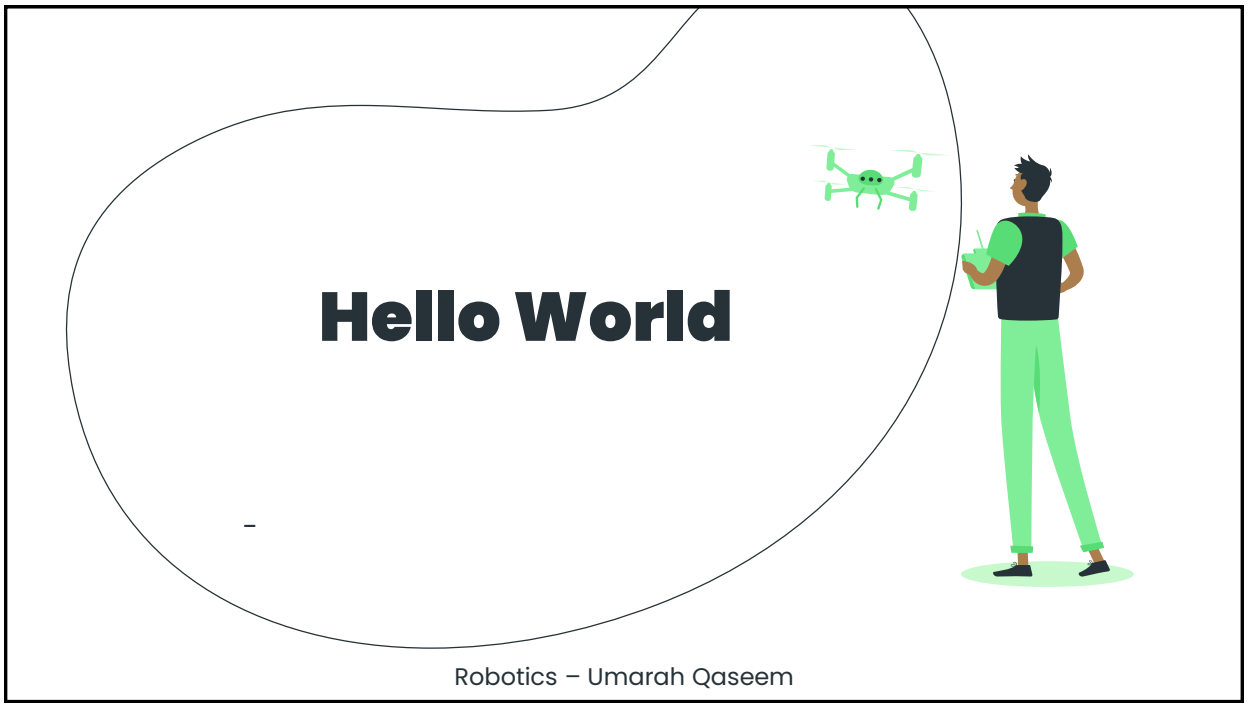
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Setup and Configurations

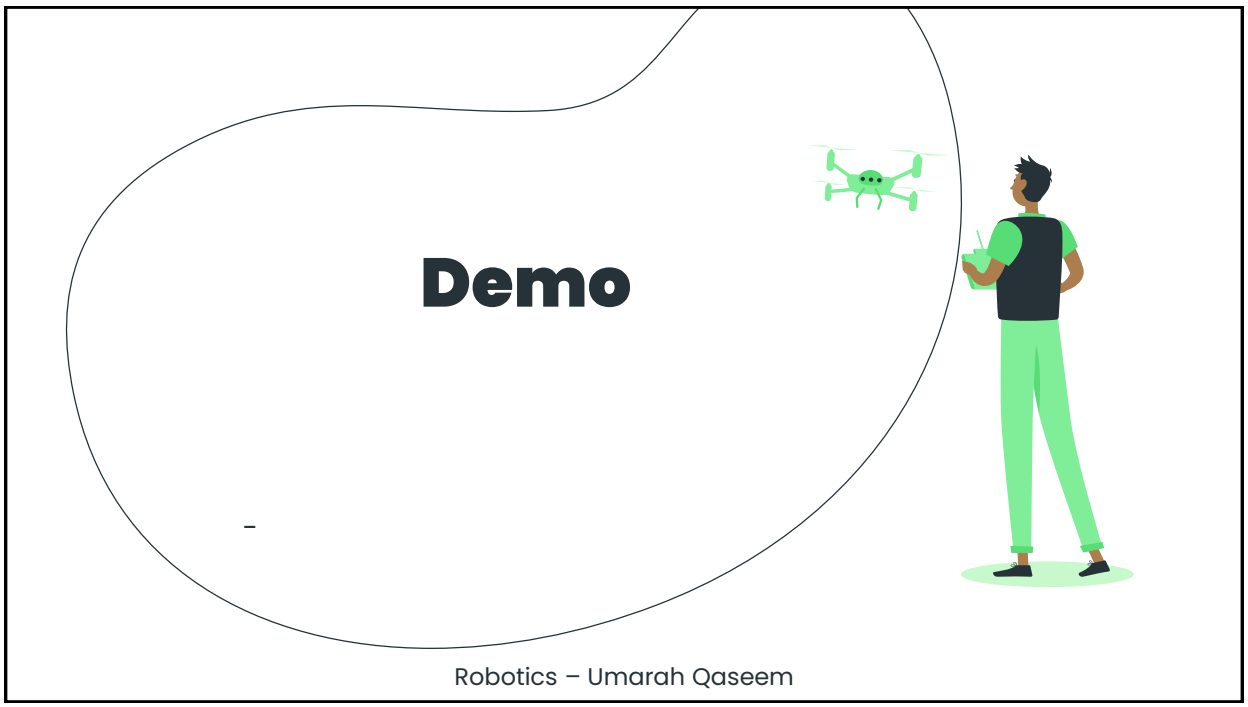


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Hello World
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Demo
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Thanks!

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